Single-cell RNA-seq of clear cell renal cell carcinoma

Tumor cells vs normal proximal tubule. Short summary. Public data from GEO (GSE159115). Full code is in the notebook scrna_seq_en.ipynb.

Question and data

Clear cell renal cell carcinoma (ccRCC) grows from cells of the proximal tubule in the kidney. I wanted to see which genes and pathways change when a normal proximal tubule cell becomes a tumor cell. I used 10x single-cell data from 7 ccRCC samples and 6 normal kidney samples (8 patients, 4 of them with both tumor and normal tissue). There were 29,499 cells before filtering and 27,166 after QC and doublet removal.

What I did

- Loaded and merged the 13 samples with Scanpy. QC: at least 300 genes per cell, mitochondrial fraction up to 25% in tumor and up to 80% in normal tissue.
- The 80% limit is on purpose. At 50% I lost 577 normal tubule cells that the authors of the dataset had kept, because these cells naturally have many mitochondria. With the final thresholds all 26,894 cells from the authors' filtered annotation stayed.
- Removed 292 doublets (Scrublet), normalized, selected 3,000 variable genes, ran PCA, corrected sample effects with Harmony, then UMAP and Leiden clustering (22 clusters).
- Named clusters by marker genes and checked them against the authors' cell labels. 17 of 22 clusters are at least 75% one authors' cell type.
- 1,351 cells from normal samples had been labeled Tumor, mostly in clusters 0, 3 and 5. The authors call most of them proximal tubule, so I relabeled them PT. After that no tumor cell comes from a normal sample.
- Differential expression on pseudobulk (PyDESeq2): counts summed per patient inside one cell type, 7 tumor patients vs 5 normal patients. Cells from one patient are not independent, so the patient is the unit, not the cell.
- Checked the result on the 4 matched patients with patient as a covariate, then ran GSEA (Hallmark gene sets) on both.

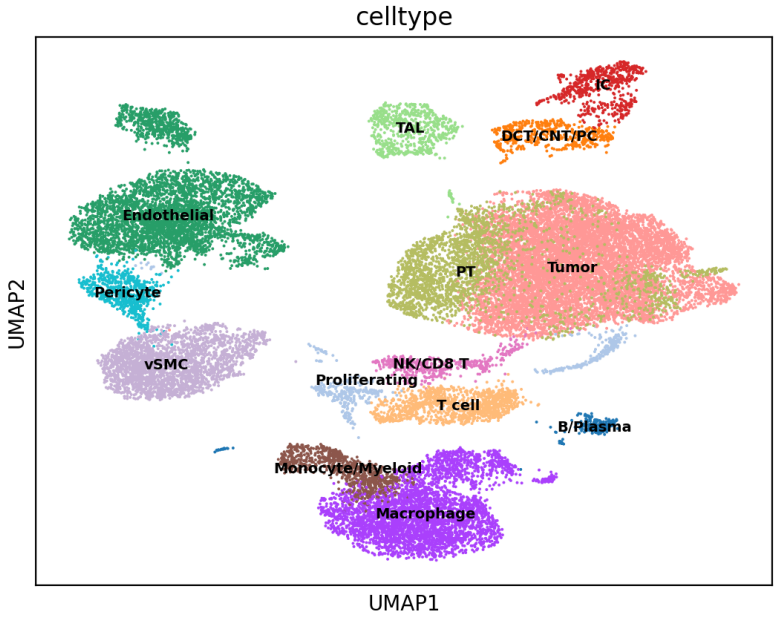


Figure 1. UMAP of all 27,166 cells colored by cell type. Tumor cells form the largest group, normal tubule parts (PT, TAL, IC, DCT/CNT/PC) appear mostly in normal samples.

Results: differential expression

1,936 genes are significant (adjusted p below 0.05). With a 2-fold cutoff, 868 are higher and 934 are lower in tumor cells. Highest in tumor: NDUFA4L2, EGLN3, NNMT, HILPDA, ANGPTL4, CA9. Lowest: ALB, ALDOB, GPC3. These are known ccRCC genes, tied to the loss of VHL and the hypoxia response. The matched-patient analysis gives 893 significant genes, 736 of them shared with the main analysis, and the fold changes correlate at 0.92.

Results: pathways

23 of 50 Hallmark gene sets are significant (FDR below 0.05). Higher in tumor: Hypoxia (NES 2.53), Glycolysis (2.48), interferon gamma and alpha response, EMT, Myc targets, mTORC1, unfolded protein response. Lower in tumor: Apical Surface, Fatty Acid Metabolism, Oxidative Phosphorylation, Xenobiotic and Bile Acid Metabolism. All 23 sets have the same direction in the matched-patient analysis, and 16 are significant there too. The four metabolic sets keep their direction but are not significant with only 4 pairs.

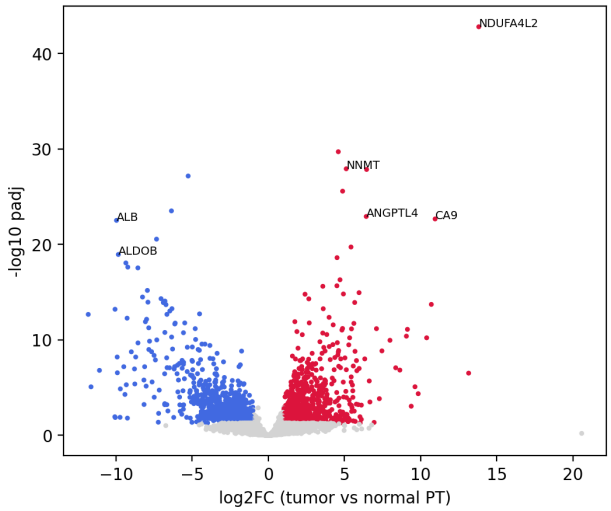


Figure 2. Tumor vs normal PT. Red: higher in tumor, blue: lower (adjusted p below 0.05 and at least 2-fold).

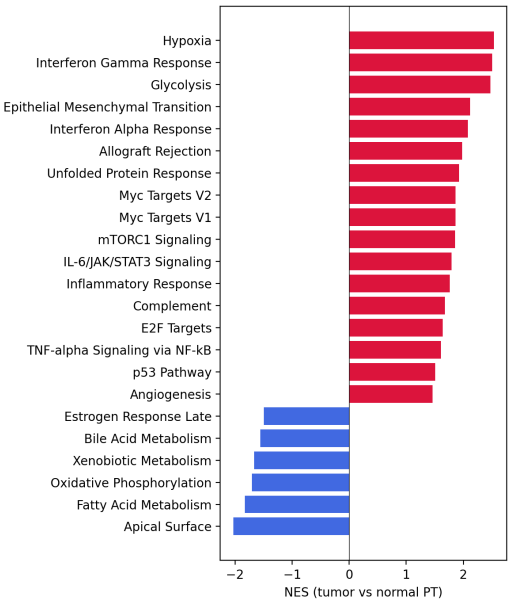


Figure 3. Significant Hallmark sets. Positive NES means higher in tumor.

In short

Tumor cells move away from the normal metabolism of the tubule (fat and oxidative metabolism, apical surface) towards a hypoxia and glycolysis program with strong growth signals. This fits what is known about ccRCC and holds in the matched-patient check.

Limits

- Small sample: 7 vs 5 patients, and only 4 pairs in the matched check. Results agree between the two, but power is limited.
- Tumor cells and normal proximal tubule look alike, so the relabeling of some cells was a manual correction based on sample origin and the authors' labels.
- Low levels of genes from other tubule parts (for example UMOD) appear in normal PT. Probably ambient RNA, but I did not test it.
- Immune signals such as interferon response may partly come from neighboring immune cells.

Data: Zhang Y et al., single cell analysis of renal cell cancers, PubMed ID 34099557.